

KOYTENDAG, Turkmenistan

WMO: 389150

Lat: 37.517N

Lon: 66.017E

Elev: 869

StdP: 14.24

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.4	25.5	12.0	10.4	40.8	16.8	13.2	43.8	28.3	58.7	25.9	58.2	3.5	270	0.455

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.5	107.2	68.0	104.8	67.7	102.5	67.2	73.3	95.7	71.0	96.5	69.5	96.3	9.0	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.1	99.5	85.5	62.6	87.8	81.6	60.0	79.8	80.1	37.5	96.0	35.4	96.2	34.1	96.5	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.9	21.8	18.3	16.0	112.0	8.3	2.1	10.0	113.5	5.2	114.7	0.5	115.9	-5.6	117.5
			15.5	75.7	8.1	4.2	9.7	78.8	5.0	81.2	0.5	83.6	-5.4	86.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	66.5	42.8	46.5	57.4	68.5	79.2	87.4	89.9	86.1	76.6	65.3	52.8	44.3	(6)
(7)		DBStd	18.42	9.66	10.16	9.59	8.76	6.56	4.83	4.16	4.72	5.67	7.71	9.00	8.45	(7)
(8)		HDD50	753	256	167	39	3	0	0	0	0	0	3	70	215	(8)
(9)		HDD65	2659	687	518	273	65	3	0	0	0	2	91	379	641	(9)
(10)		CDD50	6777	34	69	268	558	904	1123	1237	1118	797	477	153	39	(10)
(11)		CDD65	3211	0	1	37	171	442	673	772	653	349	100	12	1	(11)
(12)		CDH74	45735	0	8	338	1736	5591	10407	12393	9684	4324	1129	123	2	(12)
(13)		CDH80	27559	0	0	81	685	2988	6697	8338	6151	2213	377	29	0	(13)
(14)	Wind	WSAvg	6.3	6.2	7.5	7.8	6.8	6.4	6.9	7.0	6.1	5.1	5.1	5.0	5.4	(14)
(15)	Precipitation	PrecAvg	7.6	1.3	1.3	1.6	1.2	0.6	0.0	0.0	0.0	0.0	0.1	0.8	1.1	(15)
(16)		PrecMax	11.8	4.8	2.6	2.7	5.4	2.0	0.3	0.0	0.0	0.2	0.5	2.5	3.3	(16)
(17)		PrecMin	4.1	0.2	0.3	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	(17)
(18)		PrecStd	2.0	0.9	0.6	0.7	1.1	0.6	0.1	0.0	0.0	0.0	0.1	0.7	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.0	74.4	87.2	96.1	103.0	109.0	111.3	108.8	101.0	91.7	84.3	71.4	(19)
(20)		0.4%	MCWB	50.1	53.7	59.8	65.9	67.6	68.5	69.2	69.0	66.4	62.5	58.0	51.2	(20)
(21)		2%	DB	64.8	69.4	82.0	91.1	99.4	105.9	108.2	105.1	97.7	87.8	77.3	66.8	(21)
(22)		2%	MCWB	48.0	51.3	57.3	63.9	66.1	67.8	68.1	68.7	65.8	60.6	55.9	49.0	(22)
(23)		5%	DB	60.7	65.5	77.8	87.2	96.5	103.4	105.6	102.5	94.6	84.4	72.1	62.5	(23)
(24)		5%	MCWB	46.5	49.8	55.9	62.6	65.6	66.8	67.8	67.8	65.0	60.3	53.8	48.0	(24)
(25)		10%	DB	56.9	61.9	73.5	83.7	93.4	100.9	103.2	99.9	91.3	80.5	68.0	58.1	(25)
(26)		10%	MCWB	45.5	48.3	54.4	61.4	64.6	66.4	67.3	66.9	63.6	58.8	52.8	46.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.6	56.0	62.0	71.0	75.1	73.8	75.1	77.7	74.9	68.5	60.1	53.8	(27)
(28)		0.4%	MCD B	62.9	67.9	77.0	86.6	92.1	101.9	99.4	98.2	88.9	79.6	77.4	63.5	(28)
(29)		2%	WB	49.7	53.6	59.5	67.1	70.2	70.9	71.9	72.9	69.1	63.5	57.7	51.1	(29)
(30)		2%	MCD B	61.1	65.1	76.2	83.8	90.8	99.0	99.9	95.2	88.7	82.4	72.5	63.0	(30)
(31)		5%	WB	48.0	51.6	57.7	64.9	67.8	69.2	70.2	70.1	66.3	61.1	55.6	49.0	(31)
(32)		5%	MCD B	58.4	61.9	73.1	82.0	89.6	97.5	99.3	96.6	90.5	80.7	68.5	59.8	(32)
(33)		10%	WB	46.2	49.7	55.9	63.0	66.2	67.9	69.0	68.2	64.5	59.4	53.8	47.0	(33)
(34)		10%	MCD B	55.7	59.8	70.0	80.2	87.8	95.9	98.2	96.3	89.3	78.6	66.8	56.4	(34)
(35)	Mean Daily Temperature Range		MDBR	15.8	16.8	18.6	20.4	22.8	24.6	25.5	27.2	27.6	24.6	19.6	16.9	(35)
(36)		5% DB	MCD BR	21.5	21.2	23.1	24.9	26.5	27.7	28.2	29.4	30.6	28.2	25.8	23.6	(36)
(37)		5% DB	MCWBR	10.2	9.3	8.8	9.4	9.2	8.9	8.3	9.3	10.8	11.3	11.2	11.0	(37)
(38)		5% WB	MCD BR	18.8	18.0	20.8	22.3	23.6	25.5	26.3	27.9	29.2	26.3	22.4	19.9	(38)
(39)		5% WB	MCWBR	10.1	8.9	9.2	10.0	10.2	9.8	9.8	10.6	12.6	12.0	11.1	10.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.393	0.441	0.500	0.513	0.509	0.507	0.492	0.465	0.443	0.477	0.452	0.399	(40)	
(41)		taud	2.091	1.973	1.854	1.864	1.895	1.889	1.922	1.982	2.002	1.893	1.941	2.064	(41)	
(42)		Ebn at Noon	239	242	240	246	250	250	252	256	254	226	213	225	(42)	
(43)		Edh at Noon	38	49	60	63	62	62	60	55	51	52	44	37	(43)	
(44)	All-Sky Solar Radiation	RadAvg	692	955	1361	1817	2273	2572	2513	2290	1915	1362	858	652	(44)	
(45)		RadStd	73	110	115	159	89	47	40	49	40	80	81	74	(45)	

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	-1.49	-2.57	N/A	N/A	N/A	N/A	(46)
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	-0.54	-0.85	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page